



Python For & While Loops

Data with Dylan | Guided Notes | Python Fundamentals #3

Name: _____

Date: _____

☒ **HOW TO USE THESE NOTES:** Watch the video and fill in the blanks as Dylan explains each concept. Use the writing boxes to capture your own notes and examples. Complete the practice problem at the end!

Section 1: Introduction to Loops

Topic: Repeating code in Python

What is a loop?

Complete the definition:

A loop is a programming structure that repeats a block of _____.

An iteration is one complete _____ through a loop.

A for loop repeats a block of code a _____ number of times.

A while loop repeats a block of code while a _____ remains True.

Choosing the right loop

A for loop is usually useful when you know _____.

A while loop is usually useful when you want code to keep running until _____.

In your own words - what is the main difference between a for loop and a while loop?

Section 2: For Loops & range()

Topic: Repeating code a known number of times

Basic for loop syntax

Complete the code below:

```
for i in _____(5):  
    print(_____)
```

What values will this code print?

How does range() work?

When range() receives only one number, Python starts at _____.

The ending number in range() is _____ included.

range(5) produces the values from _____ through _____.

range(1, 11) produces the values from _____ through _____.

In your own words - explain the inclusive/exclusive rule of range(start, stop):

Loop Syntax Reference

CODE	OPERATOR/FUNCTION NAME	RESULT / NOTES
for i in range(5):	_____	Repeats using values from _____ through _____.
range(1, 11)	_____	The start value is _____; the stop value is _____.
print(i)	_____	Displays the current _____.

Circle the correctly formatted for loop

Choice A

```
for i in range(5):  
    print(i)
```

Choice B

```
for i in range(5)  
print(i)
```

Why is the other choice incorrect?

Section 3: Simple Multiplication Table

Topic: Using a for loop to repeat a calculation

Program goal

Create a multiplication table for a given number.

```
num = 5
```

Complete the loop below so it prints multiples from $5 * 1$ through $5 * 10$:

```
for i in range(_____, _____):  
    print(num * i)
```

Why does the stop value need to be 11 instead of 10?

Making output easier to read

Complete the f-string below:

```
print(f"{_____} * {_____} = {_____}")
```

The curly braces {} inside an f-string let you place _____ inside text.

In your own words - why is the f-string output more helpful than printing only the answer?

Section 4: Nested For Loops

Topic: Repeating a loop inside another loop

What is a nested loop?

Complete the definition:

A nested loop is a loop placed _____ another loop.
The loop on the outside is called the _____ loop.
The loop inside it is called the _____ loop.

Building a full multiplication table

```
for i in range(1, 11):
    print(f"Multiples of {i}")

    for j in range(1, 11):
        print(f"{i} * {j} = {i * j}")

    print()
```

When i is equal to 1, the inner loop makes j go from _____ through _____.
After the inner loop finishes all of its j values, Python returns to the _____ loop.
The next outer-loop value after i = 1 is i = _____.

Trace the loop

OUTER LOOP VALUE (i)	INNER LOOP VALUES (j)	WHAT THE PROGRAM PRINTS
1	_____ through _____	Multiples of _____
2	_____ through _____	Multiples of _____
3	_____ through _____	Multiples of _____

In your own words - what happens after the inner loop completes its final iteration?

Section 5: While Loops & Infinite Loops

Topic: Repeating code while a condition remains True

Basic while loop syntax

```
i = 0

while i < 5:
    print(i)
    i += 1
```

Complete the definition:

A looping variable is a variable that helps track _____ in a loop.
A while loop needs a looping variable declared _____ the loop.
The looping variable must change during each _____.
An infinite loop happens when a loop condition never becomes _____.

While Loop Reference

CODE	OPERATOR/FUNCTION NAME	RESULT / NOTES
while i < 5:	_____	Repeats while the condition is _____.
i += 1	_____	Increases i by _____ each iteration.
break	_____	Exits the loop _____.

Find the missing line

```
i = 0

while i < 5:
    print(i)
```

What line needs to be added to prevent an infinite loop?

In your own words - why must a while-loop variable be incremented or otherwise changed?

Section 6: Revising the Number Guessing Game

Topic: Using a while loop, attempts, and break

Program goal

Give the user up to three chances to guess a randomly generated number.

```
import random

secret = random.randint(1, 10)
attempts = 1
```

Complete the loop condition:

```
while attempts _____ 3:
```

The guess input must be placed _____ the while loop so the user can make a new guess during each iteration.

Complete the game logic

```
guess = int(input("Please enter a number between 1 and 10: "))

if guess == secret:
    print("You got it right")
    _____

elif guess < secret:
    print("Your number was too low")

else:
    print("Your number was too high")

print(f"You have {3 - attempts} attempts remaining")

attempts += _____
```

Trace the game paths

CONDITION	PROGRAM RESPONSE
guess == secret	Print that the user got it right, then _____.
guess < secret	Print that the guess was too _____.
guess > secret	Print that the guess was too _____.
The user misses all three attempts	Print that the user ran out of _____.

In your own words - what does break do when the user guesses the secret number correctly?

Section 7: Putting It All Together

Topic: Loop reference table

Using what you have learned, complete the full loop reference table:

LOOP TYPE	WHEN TO USE	BASIC SYNTAX EXAMPLE	YOUR OWN EXAMPLE
For Loop	When you know how many times code should repeat.	for i in range(5):	
Nested For Loop	When one repeated task needs to run inside another repeated task.	for i in range(1, 11): for j in range(1, 11):	
While Loop	When code should repeat while a condition remains True.	while i < 5:	

Section 8: Quick Recap

Test yourself - try to complete these without looking back!

1. A loop repeats a block of _____.
2. A for loop is most useful when you know _____.
3. range(5) starts at _____ and stops before _____.
4. A while loop keeps running while its condition is _____.
5. To avoid an infinite loop, the looping variable must _____ during the loop.

Practice Problem

 **Challenge - Pause the video and try this yourself before Dylan reveals the answer!**

```
import random
secret = random.randint(1, 10)    attempts = 1
```

Goal: Build a number-guessing game that gives the user three chances to guess a secret number from 1 through 10.

- Ask the user for a new guess during each loop iteration.
- Print whether the guess is too low or too high.
- When the user guesses correctly, print "You got it right" and use break.
- Increase attempts after an incorrect guess. Reveal the secret number when they run out of attempts.

Why must guess = int(input(...)) be placed inside the while loop?

Why does the correct-guess path need a break statement?

Write your full statement below:

How did it go? What part tripped you up?

Personal Notes

Use this space to capture any additional observations, questions, or ideas from the video.

Key Terms Reference

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Loop	
For Loop	
While Loop	
Iteration	
Looping Variable	
for	
while	
range()	
Start Value	
Stop Value	
Inclusive	
Exclusive	
Nested Loop	
Outer Loop	

Key Terms Reference (continued)

Continue building your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Inner Loop	
f-string	
Curly Braces {}	
print()	
Condition	
Increment	
+=	
Infinite Loop	
break	
input()	
int()	
random	
random.randint()	
Attempts	